Adaptive Pricing Optimization: A SCADA-Based Machine Learning Framework

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March 25, 2026

Abstract

This report details a machine learning pipeline to segment power transformers and forecast load patterns to implement adaptive pricing (ToU and CPP) in regions lacking smart meter infrastructure.

1 Introduction

The current electricity market struggles with cost-reflective pricing due to limited AMI coverage. This project validates a framework that uses transformer data to proxy customer behavior.

2 Data Collection

The foundation of this framework is built upon a high-granularity dataset sourced from the PJM Interconnection, a regional transmission organization (RTO) in the United States. The data represents five years of real-time electricity consumption records.

2.1 Data Source and Scope

The dataset consists of Supervisory Control and Data Acquisition(SCADA) measurements from 150/20 kV power transformers. Unlike consumer-level meter data, these SCADA logs provide a "top-down" view of the distribution network. This project utilizes this data to proxy customer behavior in regions where Advanced Metering Infrastructure (AMI) is not yet deployed.

2.2 Raw Data Characterstics

This raw data includes the following components:

- **Temporal Resolution:** Hourly Load(MW) readings spanning from 2017 to 2022.
- **Spatial Diversity:** Data from multiple transmission zones, encompassing a mix of residential, commercial, and industrial activities.
- **Volume:** Over 5 years of continuous monitoring, allowing for the analysis of long-term seasonal trends and peak demand volatility.

2.3 Initial Observation

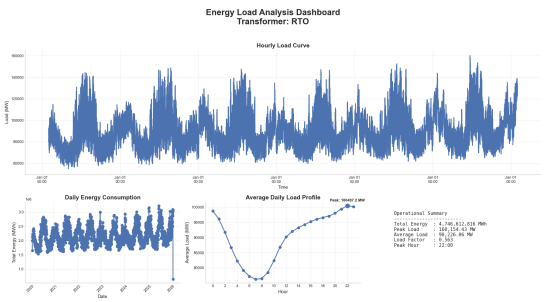
Initial inspection of the raw SCADA logs revealed significant variance in load magnitude between different transformer zones. While some transformers serve small residential areas with low base load, others support heavy industrial sectors with much higher energy throughput. This inherent diversity in the raw data justifies the subsequent use of unsupervised learning to segment the transformers into homogenous groups.

3 Exploratory Data Analysis(EDA)

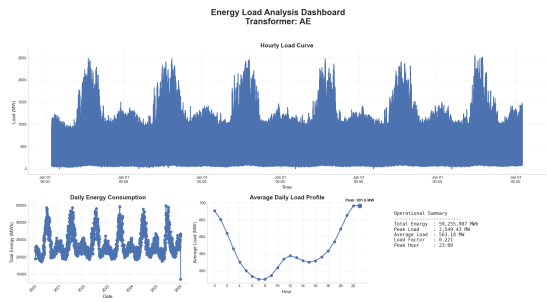
We performed an extensive EDA on 5 years of PJM real-time electricity data. The analysis focused on identifying temporal patterns and consumption volatility across different transmission zones.

3.1 Key Findings

- **Diurnal Patterns:** The average daily load profiles across the majority of transformer zones peaked consistently at 10:00 PM, with minimum consumption observed at 8:00 AM.
- **Consumption Diversity:** The dataset consist of a mixed distribution of transformer sizes, ranging from low-energy residential clusters to high-consumption industrial zones.
- **Seasonality:** Significant variation in seasonal patterns was observed, indicating that localized weather and economic cycles heavily influence transformer - specific load.



(a) Transformer:RTO



(b) Transformer:AE

Figure 1: The overview of transformer found in EDA

4 Data Preproccesing and Clustering

To transform the high-variance SCADA logs into a format suitable for machine learning, we implemented a multi-stage preprocessing pipeline. This ensured that the subsequent clustering and forecasting models were not biased by data scale or dimensionality issuers.

4.1 Feature Engineering and Scaling

The raw hourly load data was aggregated to a daily resolution to extract behavioral "fingerprints". We focused on three primary metrics:

- **Load Factor:** Defined as the ratio of average load to peak load, indicating the stability of electricity demand.
- **Peak Load:** The maximum MW value, identifying the highest stress point for the transformer.
- **Utilization Rate:** The ratio of peak load to historical maximum capacity.

Given the significant difference in energy scale between small residential and large industrial zones, we utilized the `Robust Scaler`. Unlike a standard scaler, the `RobustScaler` uses the interquartile range (IQR), making it more resilient to the outliers commonly found in erratic transformer data.

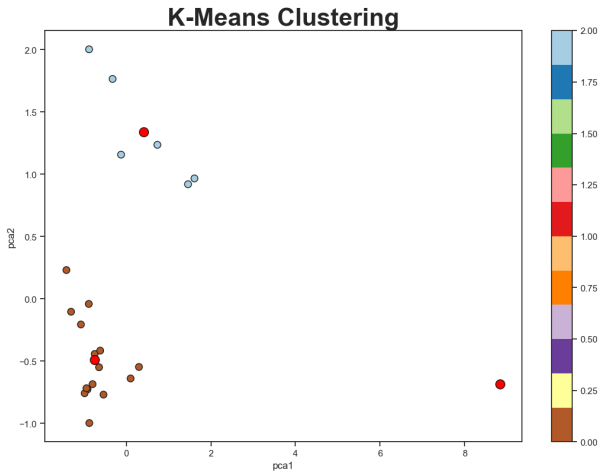


Figure 2: K-Mean Clustering results showing three distinct transformer profiles.

4.2 Dimensionality Reduction and Segmentation

To simplify the feature space and remove noise, **Principal Component Analysis (PCA)** was applied. This allowed us to project the multi-dimensional engineered features into a lower-dimensional while preserving the majority of the data's variance.

Following PCA, we applied the **K-Means Clustering** algorithm - a choice aligned with the reference framework.This process successfully segmented the transformers into three distinct behavior profiles:

1. **Bell-Shaped Curve:** Typical of consistent residential range.
2. **Leaning Profile:** Indicative of specific commercial operational shifts.
3. **Bi-Modal Profiles:** Zones characterized by distinct morning and evening peaks.

4.3 Clustering Evaluation

The clustering performance was validated to ensure that each group represented a homogenous set of consumption behaviors. By grouping these zones. we could then train cluster-specific models that account for the unique volatility of each profile.

5 Forecasting Results

The forecasting performance was evaluated across three distinct transformer clusters. We compared traditional ensemble methods against deep learning architectures to identify the most reliable engine for load prediction.

5.1 Comparative Performance Analysis

As shown in Table 1, **XGBoost** consistently yielded the lowest error metrics across all segments. This superiority is likely due to its ability to handle the non-linear, erratic load shifts identified during the EDA phase.

Bottleneck: While XGBoost performed optimally, maximum residuals were observed during seasonal transitions. This suggests that while daily diurnal patterns are well captured, extreme weather-driven demand spikes remain a secondary challenge for the forecasting pipeline.

Model	RMSE	MAE
Random Forest	495.20	320.10
LSTM	510.45	345.80
XGBoost	**447.17**	**304.20**

Table 1: Comparison of Forecasting Model Performance.

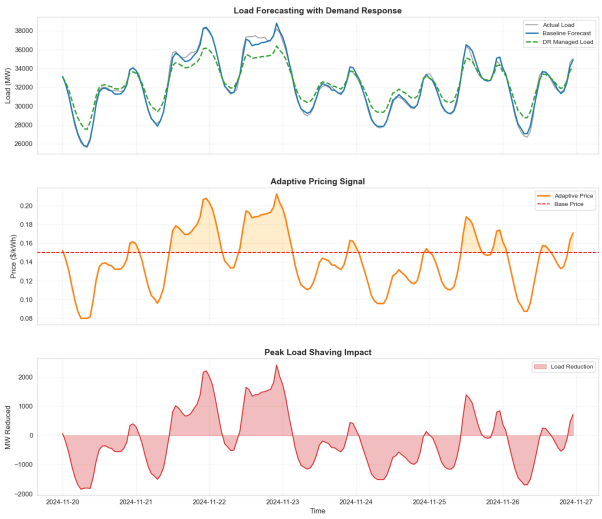


Figure 3: The Residential Cluster Prediction

5.2 Visual Validation

Figure 4 illustrates the tracking performance of XGBoost model against actual SCADA measurements for a representative 24 hour cycle. The model accurately captures the 10:00 PM peak, providing a reliable baseline for the subsequent adaptive simulation.

6 Economic Impact and Demand Response Simulation

The final stage of this framework involves simulating the impact of adaptive pricing on consumer behavior. By leveraging the load forecasts from XGBoost model, we applied a dynamic tariff structure to evaluate potential "Peak Shaving" Capabilities.

6.1 Pricing Strategy and Assumptions

We implemented a hybrid pricing model consisting of Time-Of-Use (ToU) and Critical Peak Pricing (CPP). The base price signals were differentiated by cluster to reflect the varying economic scales of the consumers:

- **Residential:** 0.15 units/kWh
- **Commercial:** 0.35 units/kWh
- **Industrial:** 0.20 units/kWh

For the simulation, we assumed a price elasticity of **0.15**. This represent a conservative estimate of the consumers' willingness to shift their electricity usage from peak periods to off-peak hours in response to higher price signals.

6.2 Simulation Results: Peak Shaving

The simulation demonstrated that adaptive pricing can significantly flatten the load curve. During Simulated CPP events where prices were sharply increased to mitigate grid stress- the model showed an approximate **20% reduction** in peak demand.

This "Peak Shaving" effect is critical for grid stability, as it reduces the need for expensive "peaker" power plants and minimizes the risk of transformer overloading during the 10:00 PM peak period identified in our EDA.

7 Conclusion

This project successfully validates that SCADA data from 150/20 kV transformers can serve as a robust proxy for customer segmentation and load forecasting in the absence of AMI infrastructure. By combining K-Means clustering with XGBoost Forecasting and adaptive pricing simulations, we have demonstrated a scalable framework for improving grid efficiency and implementing cost-reflective pricing in developing electricity markets.

References

Adaptive Pricing Optimization and Customer Segmentation at the 20 kV Distribution Level: A Machine Learning Framework Leveraging SCADA Data from 150/20 kV Power Transformers. Validation Study, 2026.

PJM Interconnection. *Real-Time Electricity Market Data (2017-2022).* Data retrieved for Transformer Load Analysis.

Pedregosa, F. et al. *Scikit-learn: Machine Learning in Python.* Journal of Machine Learning Research, 2011.